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Mechanical Performance of Three-Dimensional Printed Lattice Structures: Assembled Versus Direct Print

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Abstract

Lattice structures are composed of a collection of struts with different orientations. During slicing, the inclined struts generate multiple disjoint contours along the build direction in additive manufacturing (AM). These contours are substantially smaller in size due to the narrow cross-section of the individual lattice struts, and they can lead to contour plurality in AM processes. Contour plurality reduces the amount of continuous contact region between two successive layers, thus resulting in poor interlayer adhesion, structural integrity, and mechanical properties of the printed lattice structure. A new interlocking and assemble-based lattice structure building approach is investigated by increasing continuity in layers and avoiding support structure to minimize contour plurality. Two lattice configurations in the form of cubic and octet lattice structures are examined. The compressive performance of the designed lattice structures is compared with the traditional single-build direct three-dimensional printed lattice structures. The mechanical performance (e.g., peak stress, specific energy absorption) of the assembled structures is found to be generally better than their direct print counterparts. The empirical constants of Ashby-Gibson power law are found to be larger than their suggested values in both direct print and assembly techniques. However, their values are more compliant for octet assembled structures, which are less susceptible to manufacturing imperfections.

Keywords: lattice structure, 3D printed assembled structure, mechanical performance of 3D printed part, contour plurality

Introduction

LATTICE STRUCTURES, A CLASS of cellular solids, consist of repetitive connected members or tessellated unit cells with multiple struts that are connected through end-point contacts or nodes.¹ Assembling these unit cells in a periodic fashion forms a complex structural network,² which can be a bending- or stretch-dominated structure that follows Maxwell's criterion.³ These low-density structures have the potential to demonstrate higher performance than monolithic structures, and they are often better suited for applications in mechanical, phononic, thermal, and biological fields.^{4,5} Compared with stochastic cellular structures (i.e., irregular foams), the mechanical properties of the periodic cellular structures are more predictable and often exhibit superior performance and multifunctional properties⁶ due to their well-ordered geometries.^{7,8}

However, the far extents of three-dimensional (3D) cellular architecture are often unaffordable due to both design and

manufacturing limitations.⁹ Fabrication imperfections, that is, topological (variations in nodal connectivity and missing struts) and dimensional (variations in cellular dimensions),¹⁰ are common in lattice structures due to the multi-stage complex manufacturing process,¹ and they can cause significant degradation in their intended performance, for example, elastic moduli and compressive yield strength.^{9,11,12}

The 3D printing or additive manufacturing (AM) relies on incremental deposition of one- or two-dimensional forms of raw materials, such as polymer or metal, to form a 3D object. The AM can be used to fabricate complicated geometries, including lattice structures. However, 3D printed lattice structures are prone to deficiencies in mechanical performance, functionality, and feasibility caused by anisotropy, insufficient interlayer adhesion, micro voids or uncontrolled porosity, staircase effect, uneven surfaces, overhang support, shrinkage, and required resources.^{1,13,14}

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Cellular structures have voids designed within them by the tessellated pattern used. During slicing of the object into layers, the process can generate numerous disconnected features that are defined as contour plurality.^{13–16} The presence of contour plurality exemplifies the lack of performance by discontinuous and interrupted path plans that can cause reduced structural integrity, surface roughness, anisotropy, and weaker interlayer adhesion.^{13,17–19}

A large body of research exists in the literature aimed at comparing the characteristics and performance of lattice structures that are fabricated with extrusion-based processes. Higher compressive strength in Kagome lattice structures is reported along the print orientation and with the smoother 3D printed surface.¹⁷ Building a cellular structure in a layer-based AM usually requires a support structure. Increasing the cell's complexity or reducing its size will introduce trapped or difficult-to-remove support structures, which is another limitation for direct print lattice structures with an extrusion-based AM process.¹⁸

Dong *et al.*¹⁹ studied the significance of process parameters and build direction in extrusion-based printing of lattice structures. They used Taguchi method to optimize the process parameters for improved print quality and mechanical performance. Three non-stochastic lattice structures (BCC, BCC-Z, and 4+1 vertical strut pattern) were printed with PLA material, and their shear and bending strengths were experimentally investigated.²⁰ They found that the 3D printed lattice structures behaved well under high-shear and out-of-plane compressive load applications.

The mechanical performance of lattice structures is often predicted by using the Gibson-Ashby equation, which is expressed with a quadratic or higher-order scaling relationship between strength, modulus, and relative density. However, due to the fabrication imperfections previously discussed, the performance behavior of a 3D printed lattice may not follow the empirical constant of the power law. To address this shortcoming, post-processing of 3D printed lattice structures (e.g., annealing) has been proposed for amorphous and semi-crystalline composites.²¹

A recent attempt to delineate fundamental linkages between volumetric porosity, surface roughness defects, and the resulting performance of lattice structures was reported by Jiang *et al.*,²² where they used principal component analysis to determine the elastic strain fields variation and develop an exponential degradation factor in the Gibson-Ashby equation.

In light of the challenges associated with direct 3D printing of lattice structures, and informed by the prior works reported in the literature, a comprehensive study of lattice fabrication techniques is conducted and their performance is analyzed for extrusion-based processes. The pre-processing knowledge base of extrusion-based 3D printing is compiled and contemplated with post-processing performance measurement for lattice structure printing. The effect of contour plurality on mechanical performance is investigated.

To reduce contour plurality, a continuous filament deposition-based design is proposed that requires assembly of the separately printed parts for constructing lattice structures. Both cubic and octet lattice structures are printed by using the assembly technique. Their mechanical performance at different length scales is measured and compared with the

traditional direct print structures of similar patterns. The results from the compression tests are compared with existing analytical models for comparison.

Methodology

Design philosophy and printing of lattice structures

The 3D printing process parameters such as filament deposition direction and build direction can affect the mechanical performance of 3D printed lattices. Rajpurohit and Dave²³ showed that raster angle can be a defining factor for the tensile strength of a 3D printed part. Parts made with 0° raster angle (load along the filament deposition) exhibited the highest tensile strength as compared with all other raster angles. Kiendl and Gao²⁴ also obtained similar results, with 0° raster layout having the highest tensile strength.

Based on their deformation behaviors, cellular solids are categorized into bending- and stretch-dominated structures. The bending dominated structures typically show one or more degrees of freedom, often with sparse strut nodal connectivity. As nodal connectivity is increased, the nodes (joints) become more rigid, causing higher stiffness with struts experiencing mainly tension or compression depending on their alignment, hence leading to stretch-dominated structures.^{6,25} Both types of lattice structures are designed in this work for demonstrating the effect of continuous deposition of materials.

A set of strut assemblies, which are the building elements of 3D structures, are designed with an interlocking mechanism, as shown in Figure 1. The macro struts are printed separately on the print bed along a continuous toolpath with nearly zero support material. The relative density of both printed lattices (cubic and octet) is kept between 6% and 16% by changing the strut length from 14 to 24 mm for cubic cells, and from 9.9 to 17 mm for octet cells.

To ensure successful material deposition in each layer during direct lattice print, the cross-sectional dimensions of the struts are kept at 2×2 mm for the cubic unit cell and 1.4×1.4 mm for the octet unit cell. Below these dimensions, the continuity of extruded material becomes highly irregular. Similarly, above a 24 mm cubic cell size, the bridging method to print filament without support becomes insufficient and delamination is observed in the overhanging strut. The strut angle for the octet lattice is kept at 45° due to its support-free extrudability.

Printing parameters and processes

An extrusion-based 3D printer (Ender 3) was used to print the lattice structures in this study. All samples were printed with a nozzle size of 0.4 mm, nozzle temperature of 205°C, and bed temperature of 70°C. The digital models of the octet and cubic lattices were created in Rhino 6. A Visual Basic based script was used to generate an AM deposition plan by maximizing the continuity and connectivity discussed in our earlier work.^{15,26} Lattice structures after assembly are shown in Figure 2.

The strut assemblies were printed flat on the 3D printer bed and joined by using cyanoacrylate-based adhesive (Glue-Masters), which takes <2 min. The direct print lattice structures were printed following the bridging method for the overhanging horizontal struts. The total build times and the time saving between the cubic and octet structures using direct print and assembly manufacturing are listed in Table 1.

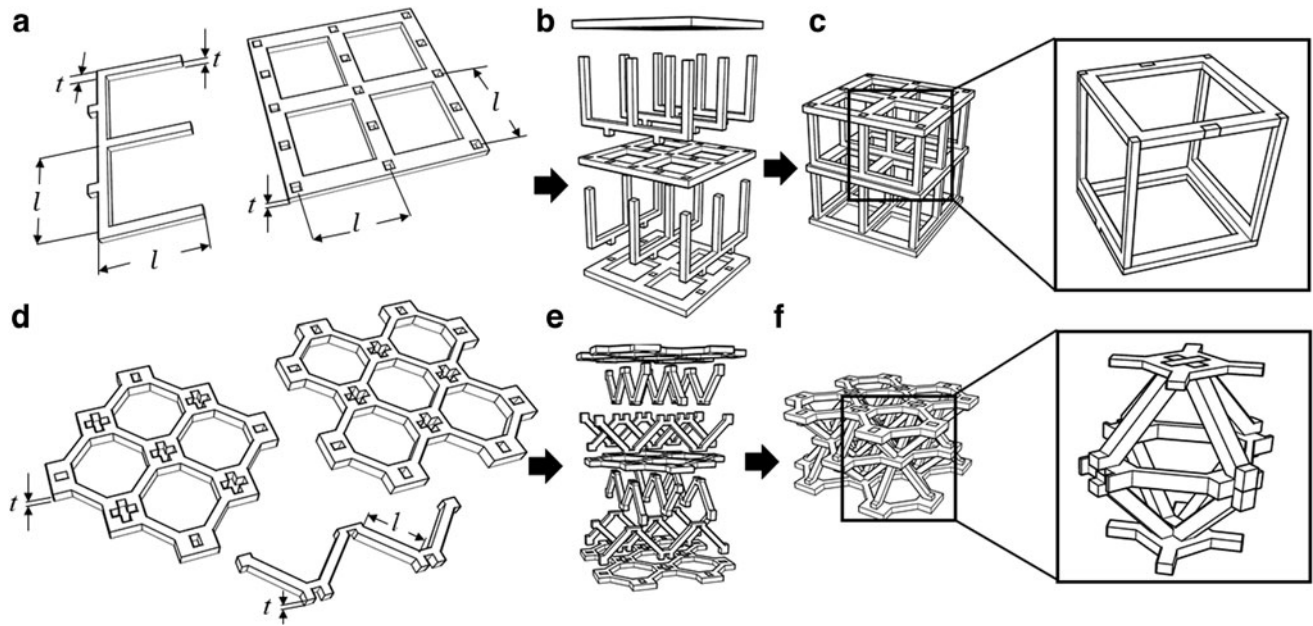


FIG. 1. (a) Unit strut sub-assembly and baseplate definition for a cubic lattice, (b) assembly method for $2 \times 2 \times 2$ cell size cubic lattice, (c) fully assembled and unit cell for cubic lattice, (d) unit strut sub-assembly and baseplates definition for $2 \times 2 \times 1$ cell size octet lattice, (e) assembly method for $2 \times 2 \times 1$ cell size octet lattice, and (f) fully assembled and unit cell for octet lattice.

Efforts were made during the design phase to minimize variation in the unit cell parameters (i.e., relative density, cell size, strut dimensions) to ensure consistency between the structure type. However, a slight variation can be observed (as shown in Table 2) due to design, extrusion, and assembly limitations.

Results and Discussion

The direct print and assembled lattice samples were tested under out-of-plane quasi-static compression by using an MTS Criterion universal testing machine with a 50 kN load cell at a 10 Hz data acquisition rate and a 5 mm/min crosshead speed. Stress values were calculated as the measured applied compressive force divided by the surface area of the lattice at each loading increment, whereas the ratios of the corresponding crosshead displacements to the initial distance

between the loading heads were used to calculate the non-dimensional displacements, commonly used strain like responses for lattice structures.^{6,27}

The compressive stress-strain curves of the cubic direct print lattice structures are shown in Figure 3a. Depending on the strut thickness-to-length ratio, t/l , as defined in Figure 1, peak stress varies from the minimum of 0.2 MPa for $t/l = 0.08$ to the maximum of 2.0 MPa for $t/l = 0.14$, with max strain reaching 0.022. If the curves were associated with a fully solid material, the transition point at peak stress would correspond to the onset of inelastic response and plastic deformation. However, for lattice structures, the transition point can be caused by different mechanisms, such as plastic deformation, lateral instability, or a combination of instability of the struts under axial compression and plastic deformation, with the latter being more confined to the highest stress locations in the joint regions.

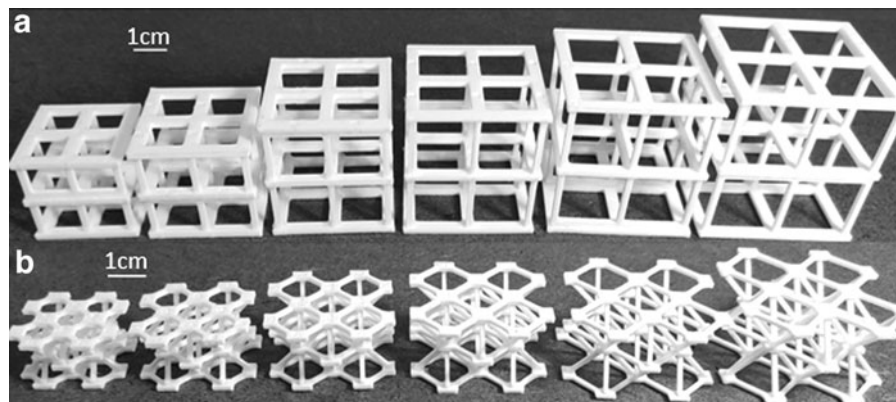


FIG. 2. Lattices designed and manufactured by the assembly method with different lattice densities. (a) Cubic lattices ($2 \times 2 \times 2$) and (b) octet lattices ($2 \times 2 \times 1$).

TABLE 1. PRINT TIME FOR CUBIC AND OCTET LATTICES

Lattice type	Cell dimension	Strut length, l (mm)	Print time for direct print lattice (min)	Print time for assembled lattice (min)	Time saving (%)
Cubic	$2 \times 2 \times 2$	14	34	21	38
		16	39	24	38
		18	49	29	40
		20	54	31	42
		22	59	34	42
		24	68	36	47
Octet	$2 \times 2 \times 1$	9.9	22	13	40
		11.31	26	15	42
		12.73	31	17	45
		14.14	35	18	48
		15.56	40	19	52
		16.97	44	21	52

The stress–strain curves of the cubic assembled lattice structures are shown in Figure 3b, with the initial peak stress ranging from 0.21 MPa for $t/l = 0.08$ to 2.33 MPa for $t/l = 0.14$. The max strain is considerably greater than those observed in Figure 3a, indicating that the structure is capable of resisting load and absorbing energy with increasing strains in the post buckled regime. Unlike in the cubic direct print structures, stress rises rapidly near the max strain, indicating a total crush and densification of the cellular structure.

For example, densification starts at $\epsilon \approx 0.8$ for $t/l = 0.08$ and gradually decreases to $\epsilon \approx 0.65$ as t/l increases to 0.14. The peak stress values for $t/l < 0.11$ are very similar in both direct print and assembled lattice structures. However, for $t/l > 0.11$, the peak stress in assembled structures surpasses that in the direct print structures.

To better examine the stress–strain response of the cubic direct print lattice structures, Figure 4a shows the photographs of the loaded structure at four distinct compressive

TABLE 2. MEASURED PROPERTIES OF DIRECT PRINT AND ASSEMBLED LATTICE STRUCTURES ($\sigma_s = 60.6$ MPa, $E_s = 2.03$ GPa)

t/l	Cubic lattice		Octet lattice	
	Direct print	Assembled	Direct print	Assembled
	Relative strength, $\frac{\sigma}{\sigma_s}(\%)$			
0.08	0.35	0.35	0.71	0.88
0.09	0.51	0.45	0.96	1.07
0.1	0.73	0.68	1.39	1.54
0.11	0.96	1.04	2.06	1.98
0.13	2.16	2.21	2.56	2.33
0.14	3.28	3.85	3.20	3.47
	Relative elastic modulus, $\frac{E}{E_s}(\%)$			
0.08	1.94	0.54	0.74	0.31
0.09	2.16	1.43	0.88	0.77
0.1	2.65	1.93	1.11	1.16
0.11	2.97	2.22	1.44	1.39
0.13	3.97	3.38	1.64	1.49
0.14	5.50	5.42	2.10	2.07
	Specific energy absorption (J/kg)			
0.08	23.8	297	1121	1358
0.09	35.7	346	1110	1845
0.1	34.9	351	1300	1859
0.11	42.9	574	1478	1792
0.13	73.7	834	2131	2593
0.14	91.6	1050	2654	2936
	Relative density, $\frac{\rho}{\rho_s}(\%)$			
0.08	6.10	5.99	5.60	5.46
0.09	7.13	7.00	6.29	6.27
0.1	8.30	8.25	7.77	7.91
0.11	9.73	9.70	9.95	9.60
0.13	12.8	12.7	12.5	12.1
0.14	15.8	16.3	15.5	16.3

Source: Liu *et al.*²⁹

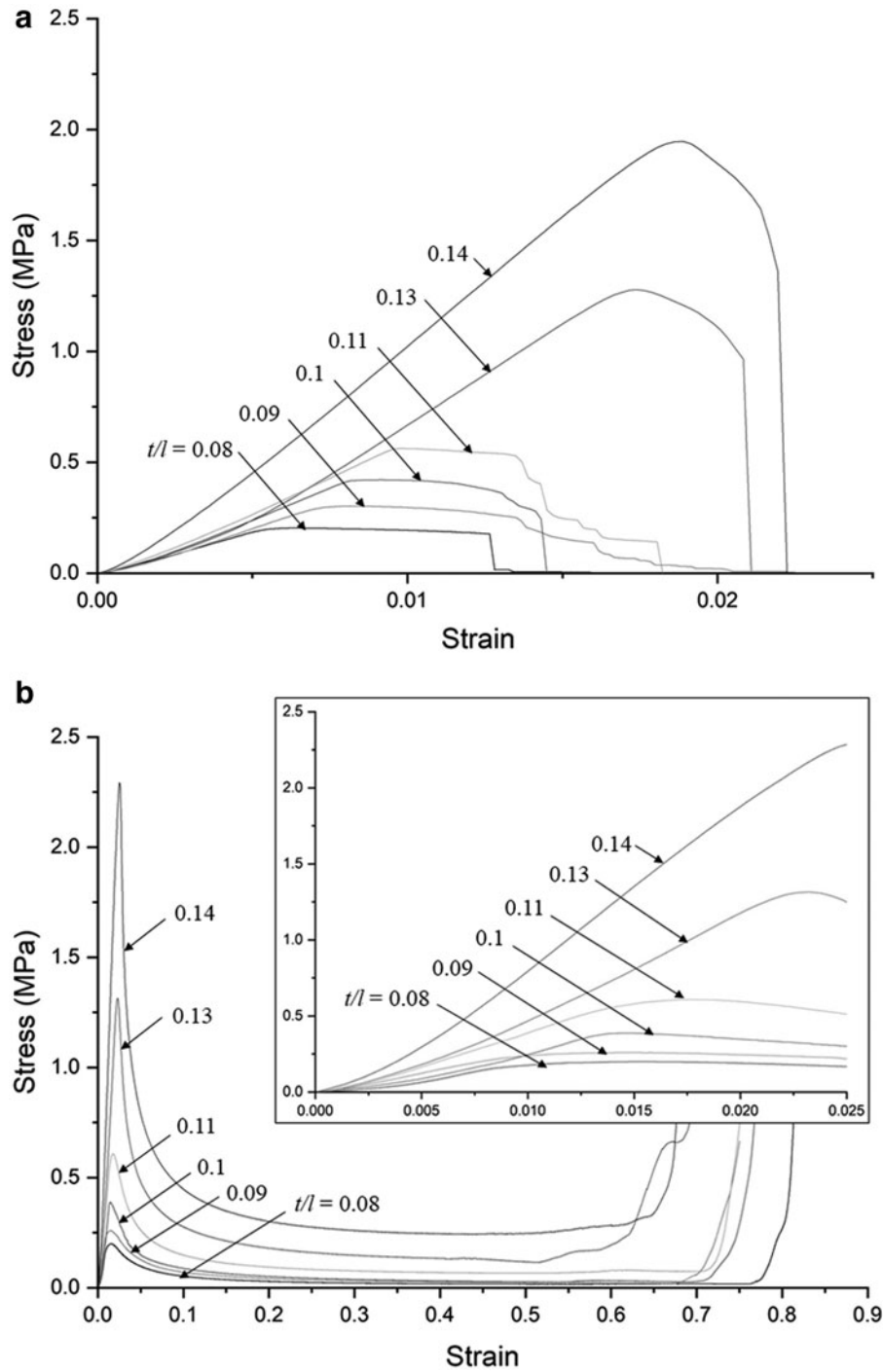


FIG. 3. Out-of-plane compressive response of (a) cubic direct print and (b) cubic assembled lattice structures at different t/l values.

strain values for the test article with $t/l = 0.11$. Lateral deformation of the struts is visible at $\epsilon = 0.009$. A closer look indicated that crack initiation began around $\epsilon = 0.014$ with rupture at $\epsilon = 0.019$ near the strut joints in multiple locations, highlighting the eventual cause of failure in these structures. It is also worth noting that failure was reached in a very short time after the quasi-static loading was initiated.

The photos in Figure 4b highlight the progressive collapse response of one cubic assembled lattice sample with $t/l = 0.11$.

Unlike the cubic direct print, crack initiation did not begin until $\epsilon \approx 0.4$ or later. A similar trend was observed in all the cubic assembled lattice structures regardless of t/l values. The plastic hinge formations observed at the nodes in Figure 4b demonstrate that the adhesive-bonded joints have no noticeable detrimental effect on the mechanical performance of the assembled lattice structure.

The differences observed in the cubic direct print and assembled lattice structures are mainly attributed to how each

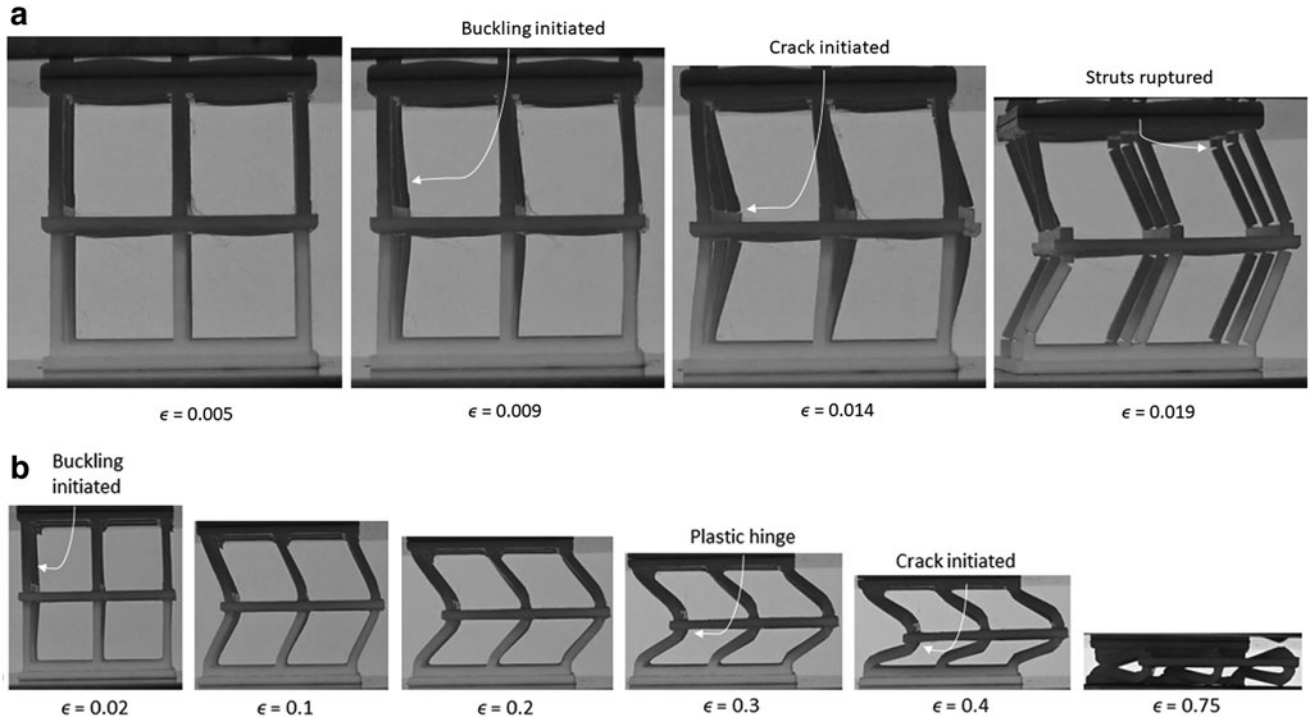


FIG. 4. Compressive behavior of (a) cubic direct print and (b) cubic assembled lattice structures for $t/l = 0.11$.

lattice is printed. In the cubic direct print lattice, the filaments are oriented in planes that are parallel to the cross-section of the struts generating contour plurality, whereas in the cubic assembled lattice, the filaments are aligned along the axis of the struts that eliminates contour plurality. When struts experience buckling-induced lateral bending, each strut cross-section experiences compression on one side and tension on the other.

The formation of cracks and rupture in the struts as seen in Figure 4a gives a clear indication of weak inter-layer adhesion aggravated by intrinsic porosity and smaller contact area between filaments in direct print struts, which ultimately result in a tensile failure.

In the case of octet lattice structures, both direct print and assembled specimens demonstrate a fairly similar compressive response at different t/l values as captured by the stress-strain curves in Figure 5 and the photos in Figure 6. The initial peak stress occurs at lower strain levels, as density is reduced for both cases. For example, at $t/l = 0.08$, the initial peak stress appears at 3% strain in direct print and 4.5% strain in assembled lattice.

The pattern is consistent with that found in the recent work that simulates the peak stress for $t/l = 0.02$ to occur at 0.4% strain for octahedral truss unit cell.²⁸ The presence of the second peak around $\epsilon \approx 0.4$ in Figure 5, not seen in Figure 3, is indicative of the formation of additional plastic hinges and a progressive collapse of the octet lattice structures, which is also not hampered in any discernable way by the use of bonded joints in the octet assembled lattice structures. A nearly total crush and densification is observed in Figures 5 at $\epsilon \approx 0.55$ for $t/l = 0.14$ increasing to $\epsilon \approx 0.8$ for $t/l = 0.08$.

Moreover, there is no evidence of delamination or fracture failure in any of the octet lattice structures before reaching

$\epsilon \approx 0.4$ regardless of the fabrication method. Contour plurality still exists in octet direct print lattice; however, the contact area between filaments has increased due to the inclined struts.

As to the reasons for the differences previously observed in the cubic direct print and assembled lattice structures not being present in the octet direct print and assembled structures, we should look at the architecture of the struts in the cubic and octet structures. The struts in the cubic lattices are longer and have a larger cross-sectional area than those in the octet lattices, but the aspect ratios are roughly equal for similar t/l values. However, as shown in Figure 1, there are 12 struts between 2 adjacent horizontal layers in the cubic lattice, but there are twice as many (24) struts in the octet lattice structures.

Also, although the struts are vertical in the cubic lattice, they are oriented at 45° in the octet lattice structures. Lastly, although in the cubic lattice structures only two vertical struts are connected at each node from top and bottom, the strut connectivity in the octet lattice structure can be in one of two forms as highlighted in Figure 1. Besides the architectural differences, there is a clear difference in how the applied compressive load is transferred to the lattice structure.

A closer examination of the behavior in Figure 4 shows that the middle horizontal layer in the cubic lattice structures experiences no discernable out-of-plane displacement under loading. However, that is clearly not the case in the octet lattice structures (Fig. 6) where the middle layer has to deform laterally to comply with the instability and lateral deformation of the struts with increasing load values, which is quite visible at $\epsilon \approx 0.1$.

What all of these differences point to is that the way the struts are loaded in the octet lattice structures makes their

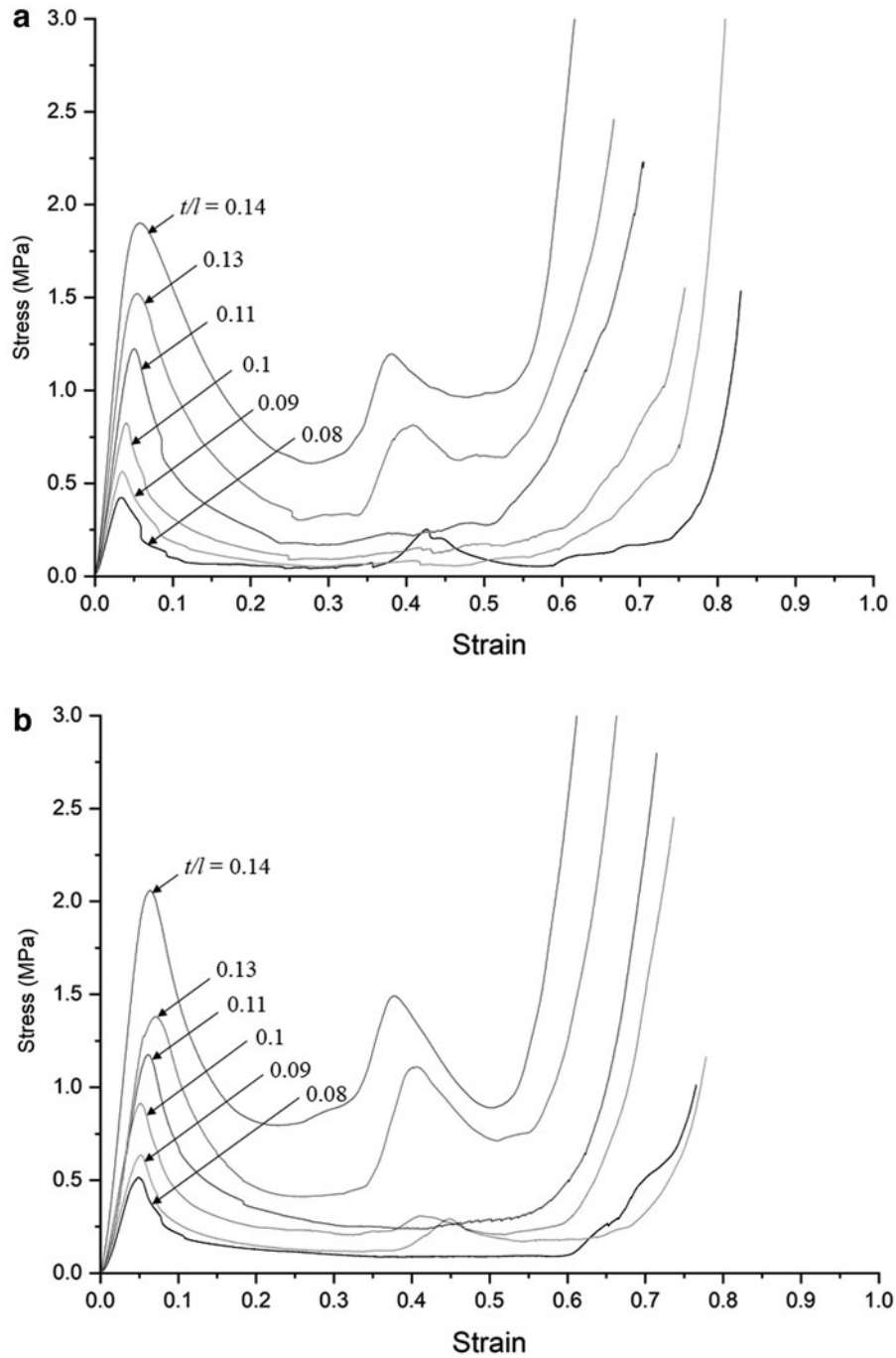


FIG. 5. Out-of-plane compressive response of (a) octet direct print and (b) octet assembled lattice structures at different t/l values.

compressive response less sensitive to the filament orientation and inherent imperfections present in each printed layer, which is not the case for the cubic lattice structures.

The plots of specific energy absorption (SEA) for the cubic and octet lattice structures are shown and compared in Figure 7 as a function of t/l . The cubic assembled lattice structures outperform their direct print counterparts, with the difference being more than 10 times for all t/l values. In contrast, the difference in SEA for the assembled and direct print octet lattice structures is not as drastic but still significant, with the

former being roughly 10–20% greater for all values of t/l . Comparing the range of SEA values, by virtue of their more progressive and ductile response, the octet lattice structures can carry approximately three to four times more specific energy than their cubic counterparts. This characteristic would be very useful in applications where energy absorption is an important design criterion.

Table 2 compares the measured properties of the cubic and octet lattice structures with each other as well as in relation to the properties of the solid PLA, as reported in the literature.²⁹

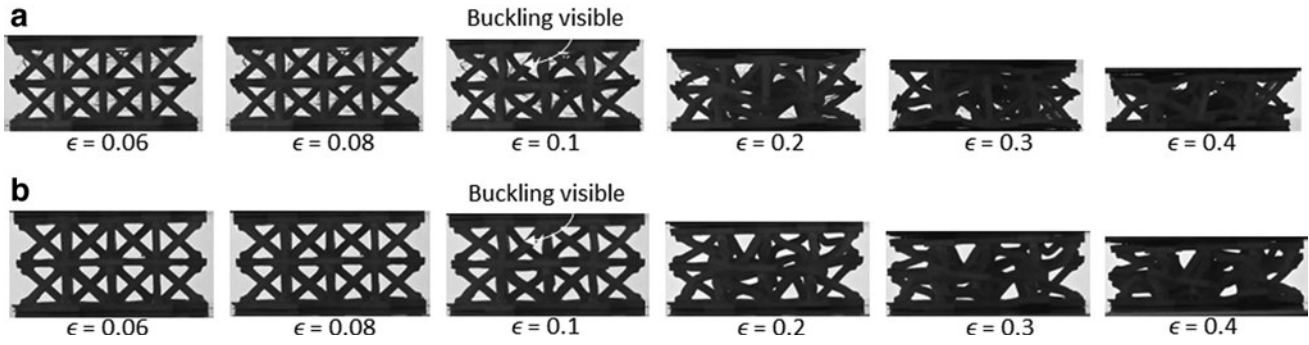


FIG. 6. Compressive behavior of (a) octet direct print and (b) octet assembled lattice structures for $t/l = 0.11$.

Since the compressive strength and elastic modulus of solid PLA are significantly greater than those of the lattice structures, the relative values shown in Table 2 for strength and modulus do not appear to vary significantly for different fabrication methods.

However, the results previously discussed show that for the cubic lattice structures, the fabrication method can make a significant difference, whereas for the octet lattice structures it does not. Similar comparisons can also be made in terms of SEA and relative density.

Prediction of strength and stiffness for octet and cubic lattice structures

The effective properties of octet lattice structures were developed by Deshpande *et al.*³⁰ using what is known as the Deshpande Fleck Ashby (DFA) model, which was further modified by Dong *et al.*²⁷ Depending on the t/l , which also affects the relative density, $\bar{\rho}$ of the struts, the peak stress in the out-of-plane (z) direction, σ_{zz} corresponding to elastic buckling and plastic yielding are estimated, respectively, as:

$$\sigma_{zz} = \frac{k^2 \pi^2 E_s \sqrt{2}}{6} \left(\frac{t}{l} \right)^4 \quad (1)$$

$$\sigma_{zz} = 2\sqrt{2} \sigma_{ys} \left(\frac{t}{l} \right)^2 \quad (2)$$

where E_s and σ_{ys} represent the elastic modulus and yield strength of the base material, respectively.

For PLA material, $E_s = 2.03$ GPa and $\sigma_{ys} = 60.6$ MPa.²⁹ Parameter k represents the column end fixity coefficient, which is bounded between 1 for pinned-pinned and 2 for clamped-clamped supports with $1 < k < 2$ representing boundary conditions that provide partial rotational restraint that fall between the two extreme cases. According to the modified DFA model, the elastic modulus of octet lattice in the out-of-plane direction, E_{ZZ} is predicted as:

$$E_{ZZ} = \frac{2\sqrt{2}E_s}{3} \left(\frac{t}{l} \right)^2 \quad (3)$$

Based on the data summarized in Table 2, the results obtained from Equations (1) and (2) are shown in Figure 8a while considering different values for k^2 . The compressive strength and modulus values obtained experimentally in this study are also shown in the same figure. The analytical equation for elastic buckling appears to capture the compressive strength of the octet lattice structures by using $k^2 = 2$ for up to $t/l = 0.11$ and $k^2 = 1$ beyond that.

Of course, as shown in Figure 5, the drop in stress after the initial peak appears to be caused by a combination of buckling and plastic deformation and that the overall response of the structure cannot be characterized by elastic buckling. The modulus predictions from Equation (3) appear to generally fall below those found experimentally, as shown in Figure 8b for most t/l values, indicating that the octet lattice structure is stiffer than the analytical prediction.

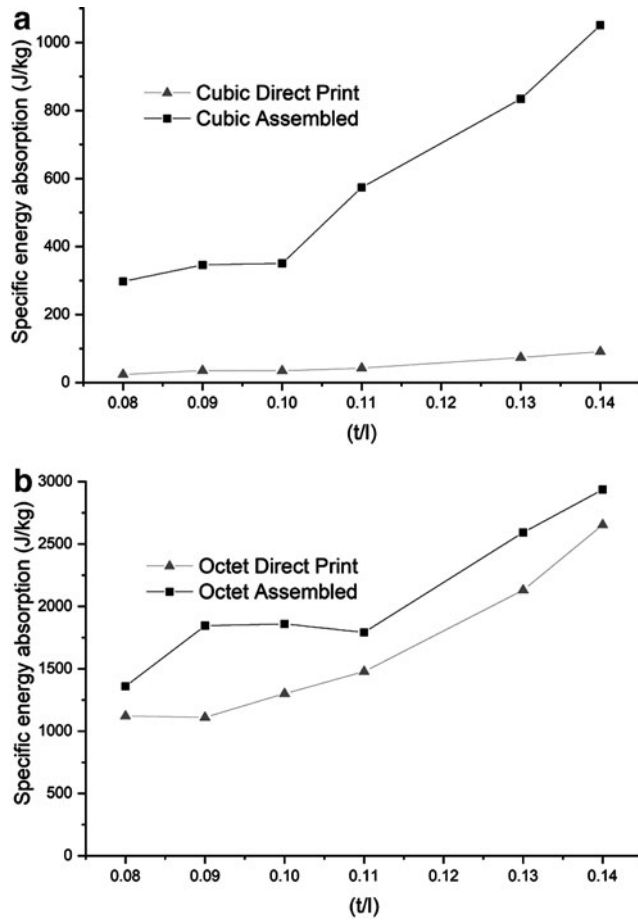


FIG. 7. Specific energy absorption at different t/l values for (a) cubic and (b) octet lattice structures with different fabrication methods.

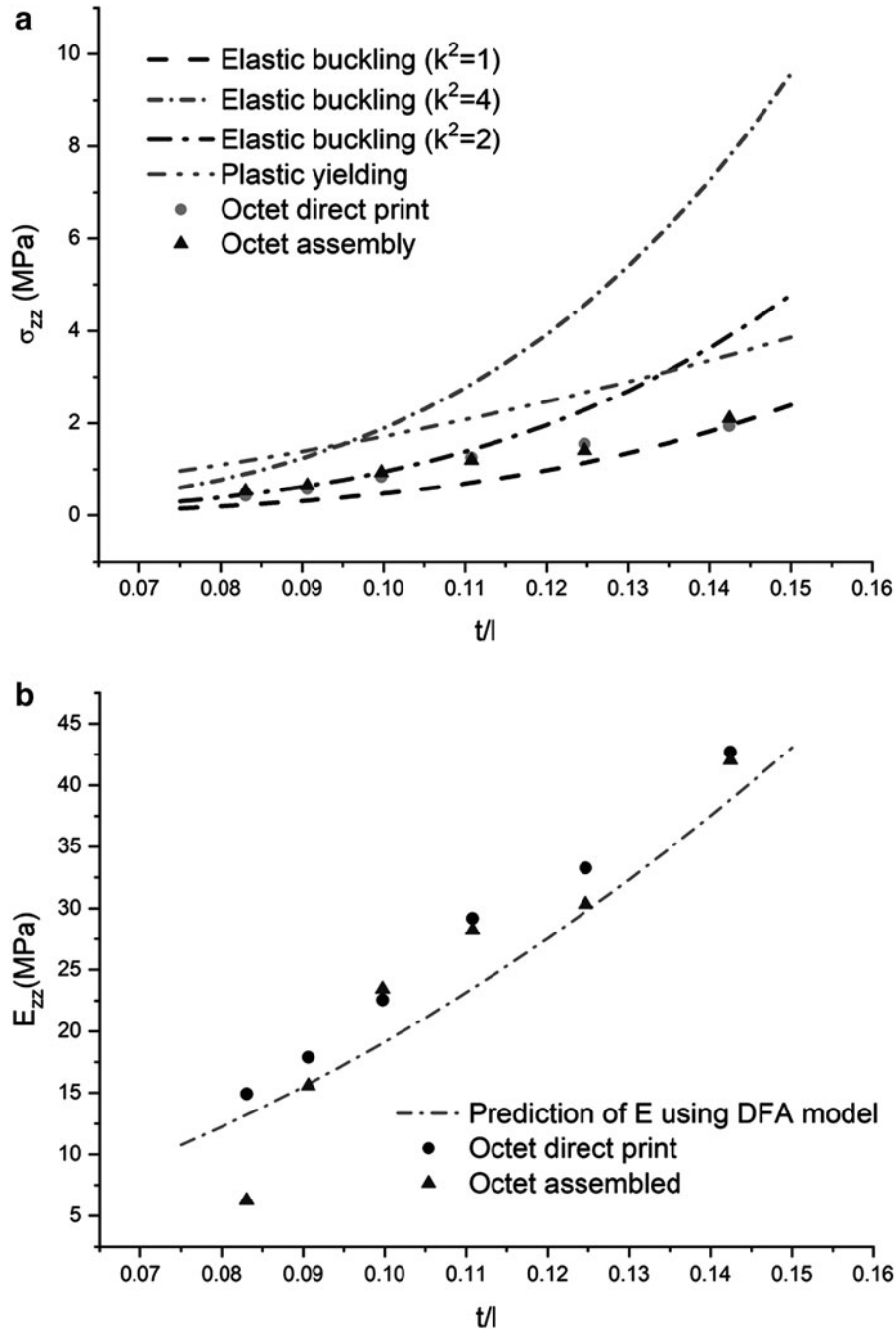


FIG. 8. Out-of-plane compressive (a) strength and (b) elastic modulus for octet lattice structures at different t/l values.

The equations for compressive responses (i.e., stress and modulus) of the cubic lattice structure are derived in a similar manner as those presented earlier for the octet lattice structures. For the $2 \times 2 \times 2$ cubic lattice structure, the peak stress, σ_{zz} corresponding to elastic buckling and plastic yielding are estimated, respectively, as:

$$\sigma_{zz} = \frac{3}{16} k^2 \pi^2 E_s \left(\frac{t}{l} \right)^4 \quad (4)$$

$$\sigma_{zz} = \frac{9}{4} \sigma_{ys} \left(\frac{t}{l} \right)^2 \quad (5)$$

The elastic modulus, E_{zz} for a $2 \times 2 \times 2$ cubic lattice structure can be predicted as:

$$E_{zz} = \frac{9}{4} E_s \left(\frac{t}{l} \right)^2 \quad (6)$$

Equations (4) through (6) are plotted in Figure 9 as a function of t/l for a $2 \times 2 \times 2$ cubic lattice structure under out-of-plane compression. The experimentally measured values of compressive failure stress and modulus are also shown for comparison. As shown in Figure 9a, at lower t/l values, the failure mode appears to be captured by the elastic buckling equation with $k^2 = 1$.

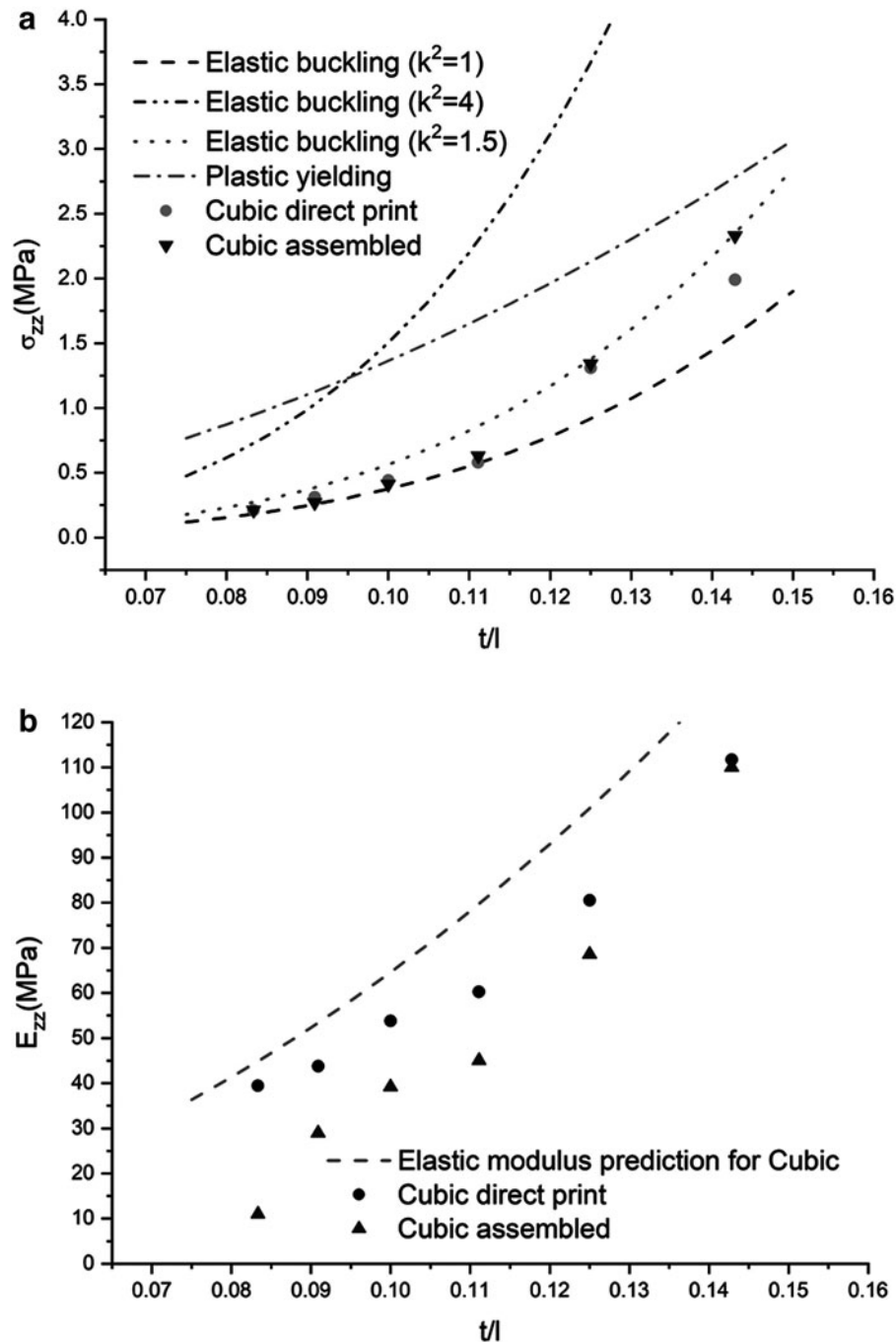


FIG. 9. Out-of-plane compressive (a) strength and (b) elastic modulus for cubic lattice structures at different t/l values.

When t/l increases, the failure mode comes closer to elastic buckling predictions with the strut joints providing partial rotational restraint with $k^2 = 1.5$ or in between the pinned-pinned and clamped-clamped boundary conditions. However, as shown by the photos in Figure 5, there is a distinct difference in compressive response of cubic direct print and assembled lattice structures, with the latter showing evidence of both buckling and plastic deformation.

This shows a different behavior than the stretch-dominated octet lattice structure. The aspect ratio of vertical struts decreases by increasing t/l , which helps the strut absorb more

load before failure. Thus, in the 3D printed lattice structure, increasing t/l improves the joints of cubic structures compared with octet structures.

The elastic modulus data is plotted for both direct print and assembled cubic structures in Figure 9b. It can be observed that the analytical equation over-predicts the elastic modulus for both direct print and assembled cubic structures. The primary reason for the difference is attributed to the fact that the struts are assumed to have a fully solid cross-section in Equation (6), whereas they tend to have some level of porosity due to inter-bead voids that are present along the entire length of each strut.

If porosity within the struts can be measured and included, the E_s value in Equation (6) would be reduced depending on the average void volume fraction among the struts, and that would tend to shift the curve downward and closer to the experimentally measured values. The fact that the measured modulus for the cubic direct print lattice structures is higher than the corresponding assembled lattice can be attributed to both higher inter-bead voids and the presence of adhesive bonds, which tend to be more flexible than PLA in the assembled lattice.

To evaluate the performance of direct print and assembled lattice structures, we analyzed their performance by using the Gibson-Ashby model of cellular solids.³¹ The performance of porous structures decreases with the topology and their relative density. This loss of performance is often expressed

with a quadratic or higher-order scaling relationship.¹ The comprehensive study by Gibson-Ashby represents the relationship between specific performance and the specific density, which is also known as a power law. Based on the failure mode analysis, we observed plastic hinges during failure. For such a failure mode, the performance loss has been expressed with the following equation:

$$\frac{\sigma_{latt}}{\sigma_{ys}} = C \left(\frac{\rho_{latt}}{\rho_s} \right)^n \quad (7)$$

Here, ρ_s is the solid material density, and ρ_{latt} is the volumetric density of the lattice structure that can be varied by changing the geometric parameters during structural design. σ_{latt} and σ_{ys} are the peak strength of the lattice and yield

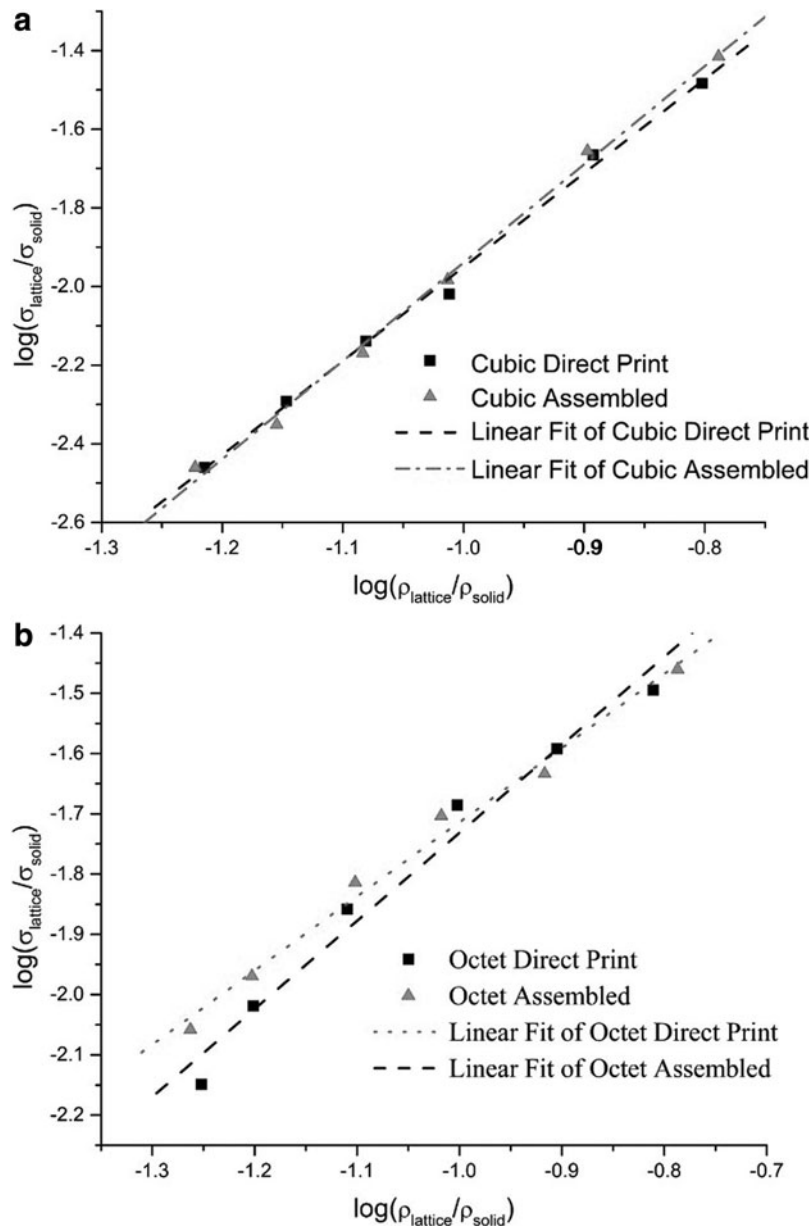


FIG. 10. Log-log plot of peak strength ratio versus density ratio for (a) cubic and (b) octet lattice structures.

TABLE 3. VALUE OF THE CONSTANTS n AND C FOR PEAK STRENGTH IN EQUATION (7)

Constant	Cubic lattice		Octet lattice	
	Direct print	Assembled	Direct print	Assembled
n	2.38	2.49	1.46	1.23
C	2.73	3.62	0.53	0.32

strength of the material, respectively. The exponential constant n indicates the loss of mechanical elastic property due to reduction in relative density, which is determined empirically. In addition, the mechanical performance loss has been attributed to constituent material properties, cell type (e.g., FCC, BCC, Cuboid),³² and arrangement of struts (i.e., number of struts joined at each node).³³

Such geometric constants of proportionality are included in the coefficient C . The expected value for bending dominated (cubic) structure is reported as $n = 1.5$ and for stretch dominated (octet) structure it is reported as $n = 1$ for plastic hinge failure.^{31,34}

To determine the empirical constants n and C , Equation (7) was plotted in a logarithmic scale (Fig. 10), and the results are provided in Table 3. A distinct difference between cubic and octet lattice can be observed in the n and C values due to their geometric load-bearing efficiency through the strut. The arrangement of the strut is often expressed via bending dominated versus stretch-dominated structure by using Maxwell's number defined as $M = b - 3j + 6$.³

Here, b is the number of struts and j the number of joints of the lattice unit cell. For $M < 0$, the structure is expected to be bending dominated or under-stiff, and otherwise the structure may be stretch dominated or over-stiff. For a bending dominated (fixed joint) cell, the value of n is often reported $= 2\sim 3$, whereas $n = 1\sim 2$ has been reported for a stretch-dominated lattice.^{22,35} The current experiment consistently matches these empirical constant value ranges for both direct print and assembled lattice structures, demonstrating the fabrication fidelity.

The empirical n and C do not account for material and manufacturing imperfections in the equation.^{9,22} It should be noted that the suggested values for the empirical constants n and C are determined assuming homogenized properties applicable to large lattice structures, where the boundary effects may be ignored. However, their experimental values shown in Table 3 are used to simply demonstrate the relative differences between the two manufacturing methods based on a common standard.

Although the direct print and assembled cubic structures have the same nodal connectivity and geometric layout, they did not perform the same due to different intrinsic defects in the two manufacturing methods. The defects in the direct print lattice structures are due to interlayer adhesion and surface roughness of the struts, including contour plurality.

The proposed assembled structures fabrication technique is designed to minimize those issues and thus demonstrate better mechanical behavior (peak stress, energy absorption). The impact of filament continuity can be observed better in the octet assembled lattice structure, which has $n = 1.23$, where the direct print counterpart has $n = 1.46$ for the loss of strength, as shown in Figure 10b and Table 3.

Conclusions

The effects of filament deposition on mechanical performance of direct print and assembled lattice structures were investigated. Both bending-dominated cubic and stretch-dominated octet lattice structures were printed by using the assembled technique and compared with traditional direct print structures. Due to the continuity of the filament deposition, both assembled lattices experienced buckling and progressive failure through formation of plastic hinges, making the assembled lattice structures suitable for energy-absorbing applications.

For the direct print lattice structures, the architecture made a difference, with the cubic lattice failing due to fracture shortly after buckling whereas the octet lattice structure continued to exhibit progressive failure.

The proposed assembly technique makes low relative density lattice structures possible ($< 6\%$), which are often not viable in the direct printing with the parameters used in this study. Similarly, the proposed technique also elevates the fabrication limitations of 3D printed lattice cell size and can print larger than 24 mm cells. This also allows the printing of smaller cell sizes, as no support structure is required for the intricate internal architecture.

Since the strut sub-assemblies are printed on horizontal orientation to ensure filament continuity, the strut angle can be varied and printed without support. This will allow the proposed assembly technique to print gradient or non-periodic lattice structures, which may not be feasible through direct printing. Currently, the assembly time is insignificant due to a relatively small number of unit cells.

However, as the number of cells is increased or the lattice itself becomes irregular, perhaps when conforming to a given shape, or when different unit cells are used in different areas, then the assembly time must be included in comparing the manufacturing efficiency of assembled with direct print lattice structures.

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